HW3: Particle Filter Tracking

Omri Mali : 212047542, Roy Ziman: 327703013

Part A

Question 1:

We first define the variables:

a. Posterior ($P(X_t \mid Z_{1:t})$): The updated probability distribution of the hidden state X_t after incorporating all available sensor measurements $Z_{1:t}$ till the current time step.

b. Prior ($P(X_t \mid Z_{1:t-1})$): The predicted probability distribution of the current state before looking at the new measurement. Computed purely by projecting the previous state forward using the system's dynamic motion model.

c. Likelihood ($P(Z_t \mid X_t)$): The probability of observing the current measurement Z_t given that the system is in a specific candidate state X_t. It evaluates how well a candidate hypothesis matches actual sensor data.

The formula $Posterior \propto \; Prior \times \; Likelihood$ defines the recursive Bayesian update loop used by both Kalman and particle filters to track an object over time. The prior represents the predicted state distribution projected forward by a dynamic motion model before seeing new data. The likelihood measures how well a candidate state matches the current sensor measurement. Multiplying them yields the posterior, which is the updated probability distribution of the hidden state given all observed data. While the Kalman filter applies this analytically to update the mean and variance of a Gaussian distribution, the particle filter calculates the prior by propagating individual particles through space and uses the likelihood to weight each particle, allowing it to model non-Gaussian, arbitrary probability shapes.

Question 2

a. We use the color histogram to score the patches because it captures the global distribution of colors within the target region, entirely discarding the rigid spatial arrangement of the pixels. This method has pros and cons:

Pros: the method provides high robustness against non-rigid geometric deformations (like a person changing posture), partial occlusions and rotations.

Cons: the method completely loses spatial structure information (meaning different spatial patterns with identical color compositions will yield the same histogram) and can easily drift if the background contains similar background colors.

b. The Sum of Squared Differences (SSD) operates directly in the image pixel space, meaning it compares exact pixel intensities at exact, rigid (x, y) coordinate alignments. The method is extremely sensitive to any spatial modifications. If the tracked object undergoes slight rotation, shape deformation, or translation, the pixel-to-pixel correspondence breaks down, causing the SSD score to increase drastically even if the patch is still tracking the correct object.

c. We can use the measurement matrix of Structural Similarity Index (SSIM): a classical method that measures the similarity between two patches based on three distinct, perceptually relevant components: luminance (brightness), contrast, and structure (the spatial pattern of pixel intensities). Instead of blindly subtracting pixel values like SSD, SSIM calculates these metrics over local windows using mean, variance, and covariance, focusing on how human vision naturally perceives structural changes in a scene. The method has the following pros and cons:

Pros: Because it normalizes for luminance and contrast, SSIM handles global lighting shifts and shadows in a really good manner. It also evaluates local pixel patterns rather than exact pixel-to-pixel alignments, making it less rigid than SSD to minor tracking inaccuracies or uniform motion shifts.

Cons: The method remains a spatial-domain template matching method. It still struggles with severe non-rigid shape deformations (like a dancing person bending over) because it expects the overall structural layout of the object to stay relatively consistent with the target template. It is also computationally heavier to calculate across 100 particles than a simple color histogram.

Question 3

For scale changes, the basic implementation provided in the assignment will struggle or fail because the state vector handles width and height as constants. If a person walks toward or away from the camera, the bounding box remains a fixed size. This causes the box to either crop out parts of the object or sample outside background noise into the histogram, destroying tracking accuracy. To fix this, scale parameters would need to be dynamically added to the state vector.

For viewpoint changes, the particle filter will handle minor to moderate viewpoint changes well because the global color histogram is spatially invariant. For example, if a person turns around slightly while dancing, the overall color composition remains roughly the same. However, if the viewpoint changes drastically (for example, exposing a side with completely different colors or clothing patterns), the similarity score against the initial template will drop significantly, which will lead to a tracking failure.

Question 4

For knowing when to update the template, we will monitor the maximum particle weight w_{max} or the similarity score of the best-fitting particle in each frame. If this score drops below a set threshold, indicating the object's appearance has drifted due to lighting changes or viewpoint shifts, it triggers an update. However, we will also set a lower threshold for freezing updates during severe drops. This will prevent the template from accidentally adapting to a passing obstacle or background clutter (occlusion drift).

For determining the new template patch, instead of replacing the template directly with the current frame's best patch, which causes rapid error accumulation and drift, we will use a running average to smoothly blend the original template (q_{orig}) with the best particle's histogram (p_{best}) from the current frame:

$$q_{new} = (1 - \alpha) * q_{prev} + \alpha \cdot p_{best}$$

Setting some blending factor α allows the tracker to adapt dynamically to environmental transformations while preserving the historical identity of the tracked object.